

StructMamPose: From Sequential Perception to Structural Reasoning for 3D Human Pose Estimation

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Abstract

Accurately modeling complex spatiotemporal dependencies and depth information is critical for monocular 3D human pose estimation. However, existing Mamba-based methods struggle to fulfill these demands, suffering from internal state update confusion induced by forced sequence flattening and depth modeling confusion inherent to single-view observations. To address this confusion, we propose a StructMamPose framework equipped with Identity Anchoring Mechanism (IAM) and View Transformation Hub (VTH). The IAM injects spatiotemporal identities into the parameter generation network to anchor the selectivity of state update matrices, suppressing spurious connections to enforce feature propagation along valid spatiotemporal dependencies. The VTH performs internal coordinate rotation to transform implicit depth inference into observable planar features, endowing the model with explicit spatial understanding and multi-view constraints. Experimental results demonstrate that our framework achieves SOTA performance on most datasets.

1. Introduction

Monocular 3D Human Pose Estimation (3D-HPE) aims to estimate 3D poses from single-view 2D observations (Hao & Li, 2025; Han et al., 2025; Li et al., 2025a). This task requires not only capturing highly complex motion dynamics but also inferring depth information (Zhang et al., 2025a; Liao et al., 2025). Such a profound human-centric understanding empowers machines to perceive human behaviors and interact with the real world, thereby facilitating widespread applications in embodied intelligence (Majumdar et al., 2023), human-computer interaction (Huo et al.,

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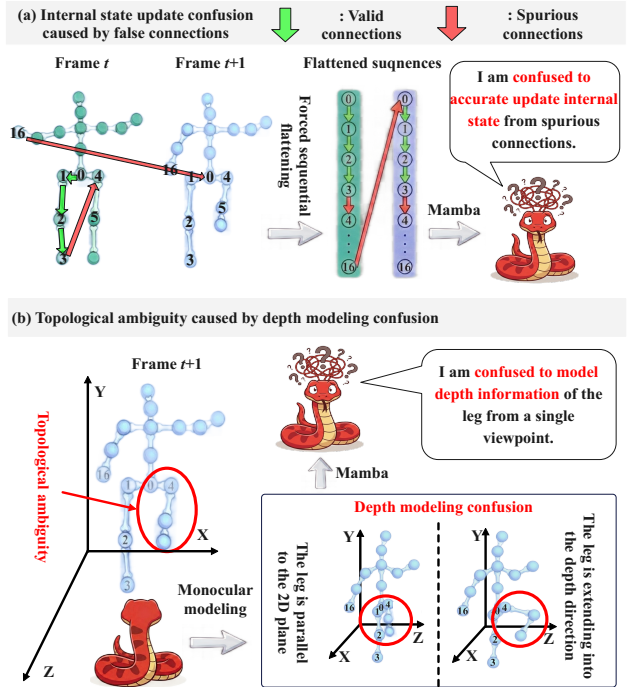


Figure 1. Illustration of our motivation: the two confusions in Mamba-based 3D-HPE. (a) Internal state update confusion stems from the forced flattening of structured topological data, which introduces spurious connections and disrupts authentic topological dependencies. (b) Depth modeling confusion arises from the inherent topological ambiguity of the monocular view, where the absence of multi-view constraints leads to depth uncertainty.

2023), and the metaverse (Zhou et al., 2023).

Recently, Mamba has made significant progress in 3D-HPE due to its superior long-range modeling capabilities and linear complexity (Yang et al., 2024; Xie et al., 2025; Li et al., 2025b; Zheng et al., 2025; Dong & Lee, 2025; Liu et al., 2025b). Specifically, PoseMamba models the human topology as a flattened sequence and performs spatiotemporal scanning, establishing a baseline for Mamba in 3D-HPE (Huang et al., 2025). Building on this, PoseMagic proposes a hybrid Mamba-GCN architecture, utilizing graph convolution network (GCN) to reinforce the capability to represent local fine-grained features (Zhang et al., 2025b). Furthermore, SAMA introduces a motion-adaptive module

to improve the hidden state modeling, which resolves the oversight of intrinsic joint motion dynamics in implicit processing (Lu et al., 2025a). Although these methods appear plausible at first glance, a deeper analysis reveals that they fundamentally suffer from severe confusion.

This confusion arises from two perspectives. First, Mamba forces the flattening of human topological data into a 1D sequence, which makes its internal state update confusing. As illustrated in Figure 1 (a), the Right Hip (Node 4) is topologically correlated with the Pelvis (Node 0) and the Right Knee (Node 5). However, forcibly flattening the 2D data into a 1D sequence disrupts the correlation of joints and introduces a spurious connection (Right Hip and Left Ankle) in Fig.1 (a). Furthermore, the inter-frame transition creates a spurious connection where the terminal node of the preceding frame immediately precedes the initial node of the subsequent frame (Node 16 of the preceding frame and Node 0 of the subsequent frame in Fig.1 (a)). Such adjacent nodes within the flattened sequence lack intrinsic causal dependency, which violates Mamba’s core assumption that ‘adjacency implies correlation’. Consequently, the model erroneously aggregates features from spatiotemporal unrelated nodes, leading to internal state update confusion.

In addition, the limitation of monocular observations inevitably causes depth modeling confusion, leading to depth ambiguity. This is because projecting a 3D scene onto a single 2D plane creates a “one-to-many” mapping where infinitely many different 3D configurations can result in the exact same 2D pose. As shown in Fig.1 (b), a single 2D leg lifting action can be interpreted as multiple 3D poses from different perspectives. Naturally, the model cannot distinguish whether the motion of the limb is parallel to the plane or extending into the depth direction from a single perspective. Therefore, the lack of explicit multi-view constraints leads to confusion in the depth modeling.

Our core insight is to reconstruct precise Mamba’s internal state update and depth modeling, endowing the model with explicit awareness of human spatiotemporal dependencies and multi-view constraints. To begin with, we introduce an Identity Anchoring Mechanism (IAM) to resolve the internal state update confusion caused by the ‘adjacency-implies-correlation’ assumption. Specifically, we inject spatiotemporal identities into both the state input matrix B and the output matrix C . Matrix B leverages the spatiotemporal attributes embedded in identities to establish topological dependencies and suppress spurious dependencies in hidden states, while matrix C selectively anchors related features based on the identity, ensuring extracted features strictly aligned with the semantics of the current joint. The combination of B and C ensures that only correlated information is propagated and retrieved, forcing the internal state to update along the actual human topological

dependencies rather than the flattened sequence. In addition, we construct a View Transformation Hub (VTH) to resolve the confusion in depth modeling. This hub performs coordinate rotations to create virtual multi-view observations in the feature space. These virtual observations not only provide explicit human spatial structure cues, but also impose multi-view constraints to rectify topological distortions from reliable frames. By incorporating such complementary human pose perspectives, this design enhances the model’s intrinsic reasoning capability for 3D human spatial structure and improves the credibility of the rectified pose. Finally, the combination of IAM and VTH equips our model with precise state update and multi-view constraints depth modeling. As a result, our framework achieves SOTA performance on the Human3.6M and MPI-INF-3DHP datasets while maintaining competitive efficiency. Ablation studies and visualizations validate the excellent performance, especially in handling depth ambiguity. The code is available at <https://github.com/JJHss6/StructMamPose>.

2. Related Work

2.1. Mamba-based Sequence Modeling in 3D-HPE

Mamba has been introduced to 3D HPE, leveraging its linear complexity to handle long motion sequences efficiently. PoseMamba replaces standard self-attention mechanisms with Mamba blocks, utilizing a four-direction scanning strategy to capture spatial topological structures and temporal node dynamics. This design not only preserves spatiotemporal correlation but also effectively maintains linear computational complexity (Huang et al., 2025). To enhance spatial representation, PoseMagic employs a dual-path architecture, using Mamba for global context and parallel GCNs for local skeletal details (Zhang et al., 2025b). HyperGCN-Mamba integrates hypergraph convolutions to explicitly extract complex skeletal geometries, followed by Mamba to model temporal limb dependencies (Cui & Hayama, 2025). Gamba proposes learning multi-level human knowledge synergistically to effectively tackle the ill-posed problem (Zhou et al., 2026). SAMA proposes a structure-aware and motion-adaptive framework to independently capture spatial joint topology and diverse motion dynamics (Lu et al., 2025b).

However, these methods forcibly flatten skeletal graphs into 1D sequences, which establishes dependencies based on ‘index adjacency’ rather than ‘topological connectivity’, leading to internal state update confusion. Since the model cannot distinguish between spurious sequence neighbors and valid topological connections, the model erroneously integrates features from unrelated nodes. To address this, we inject explicit spatiotemporal identity, which acts as navigation anchors to suppress the information propagation along spurious connections, ensuring that the internal states evolve strictly according to authentic topological correlations.

2.2. Resolving Depth Ambiguity in 3D-HPE

Recovering 3D human structure from a single 2D visualization is inherently ill-posed due to the loss of depth information (Hassan et al., 2019; Li et al., 2022; Liao et al., 2024; Hosseini et al., 2025). Existing methods mainly employ implicit priors or constraints to reduce this ambiguity. For instance, Unified 2D-3D Discrete Priors matches 2D features with valid 3D structures in a shared codebook to enforce realistic body shapes (Chen et al., 2026). PandaPose maps features to preset 3D anchors, providing spatial references to guide depth regression (Zheng et al., 2026). Focusing on dynamics, KTFFormer uses trajectory priors to ensure temporal depth consistency (Peng et al., 2024). For spatial relations, DSG-Transformer infers relative depth by analyzing semantic connections in the skeletal graph (Zhang et al., 2024). Furthermore, Ensembles on Flat Loss Landscapes (EFL) employs a redundant solution strategy to neutralize model instability for better generalization (Han et al., 2026).

Despite these efforts, these approaches rely on single-view observation and lack effective multi-view constraints to resolve spatial modeling confusion. Consequently, they struggle to accurately infer spatial depth from a single 2D perspective. This ambiguity inevitably leads to structural distortions in the estimated 3D pose. To achieve multi-view constraints without external data, we conduct coordinate rotation within the feature space. This transforms implicit depth information into explicit planar representations, facilitating structure perception via multi-view spatial cues. Furthermore, it imposes constraints and rectifications derived from reliable temporal frames and topological joints, effectively eliminating depth modeling confusion and ensuring structural rationality.

3. Method

3.1. Preliminary

State Space Models (SSMs) provide a robust framework for modeling continuous linear time-invariant (LTI) systems and analyzing causal sequences (Liu et al., 2024). These systems mathematically map a 1D function or sequence through an implicit latent state, formulated via the following differential equations:

$$\begin{aligned} \mathbf{h}'(t) &= \mathbf{A}\mathbf{h}(t) + \mathbf{B}\mathbf{x}(t), \\ \mathbf{y}(t) &= \mathbf{C}\mathbf{h}(t) + \mathbf{D}\mathbf{x}(t). \end{aligned} \quad (1)$$

In this formulation, $\mathbf{x}(t) \in \mathbb{R}^L$ represents the input signal, while $\mathbf{y}(t) \in \mathbb{R}^L$ denotes the resulting output. The core component, $\mathbf{h}(t) \in \mathbb{C}^N$, acts as a latent state variable that encapsulates historical context, effectively serving as the system’s compressed memory. The evolution of this state is driven by the state transition matrix $\mathbf{A} \in \mathbb{C}^{N \times N}$ in the

absence of external influence. Meanwhile, the input matrix $\mathbf{B} \in \mathbb{C}^{N \times 1}$ dictates how external signals are projected into the hidden state. Finally, the projection matrix $\mathbf{C} \in \mathbb{C}^N$ translates the hidden state into the output, with $\mathbf{D} \in \mathbb{C}^1$ providing a direct residual pathway. Since deep learning inputs are inherently discrete, translating the continuous SSM into a discrete recurrence is essential. Following Mamba and PoseMamba, we utilize the Zero-Order Hold (ZOH) discretization principle. This method assumes the input signal $\mathbf{x}(t)$ remains constant within a sampling interval determined by the timescale parameter Δ . This parameter Δ is crucial as it creates a bridge between continuous and discrete domains, effectively regulating the trade-off between retaining historical state information and incorporating new input. The discretization transforms the continuous parameters into discrete forms $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$ as follows:

$$\begin{aligned} \bar{\mathbf{A}} &= e^{\Delta \mathbf{A}}, \\ \bar{\mathbf{B}} &= (\Delta \mathbf{A})^{-1}(e^{\Delta \mathbf{A}} - \mathbf{I})\Delta \mathbf{B}. \end{aligned} \quad (2)$$

To transition from a static LTI system to a dynamic, content-aware model, the parameters are discretized in an input-dependent manner. Specifically, linear projections are applied to the input $\mathbf{x}(t)$ to generate time-varying parameters $\Delta_t = s_\Delta(\mathbf{x}(t))$, $\bar{\mathbf{B}}_t = s_{\bar{\mathbf{B}}}(\mathbf{x}(t))$, and $\mathbf{C}_t = s_{\mathbf{C}}(\mathbf{x}(t))$, where $s(\cdot)$ represents the learnable projection layers. Consequently, the discrete-time forward pass is computed via the following recurrence:

$$\begin{aligned} \mathbf{h}_t &= \bar{\mathbf{A}}\mathbf{h}_{t-1} + \bar{\mathbf{B}}_t\mathbf{x}_t, \\ \mathbf{y}_t &= \mathbf{C}_t\mathbf{h}_t + \mathbf{D}\mathbf{x}_t. \end{aligned} \quad (3)$$

3.2. Overview of the StructMamPose Framework

We propose StructMamPose to achieve robust 3D-HPE. The overview is as illustrated in Figure 2. We first project 2D pose input into the feature space. To align with Mamba’s sequential modeling paradigm, these features are flattened into 1D sequences from four directions, and processed by the Mamba blocks. Each block is equipped with an IAM to inject spatiotemporal identities into the state update matrices, rectifying the confusion arising from spurious connections within the flattened sequence. Furthermore, we embed a VTH within the intermediate block to resolve depth modeling confusion. Specifically, the intermediate block output is decoded to obtain an intermediate 3D pose. The VTH performs a coordinate rotation on this pose to generate virtual side-view observations for all frames. By facilitating interaction between the original front and generated side views, the model explicitly perceives the human spatial structure and utilizes spatiotemporal multi-view constraints for error correction. These interacted multi-view features are then fed into subsequent blocks to enhance spatial understanding

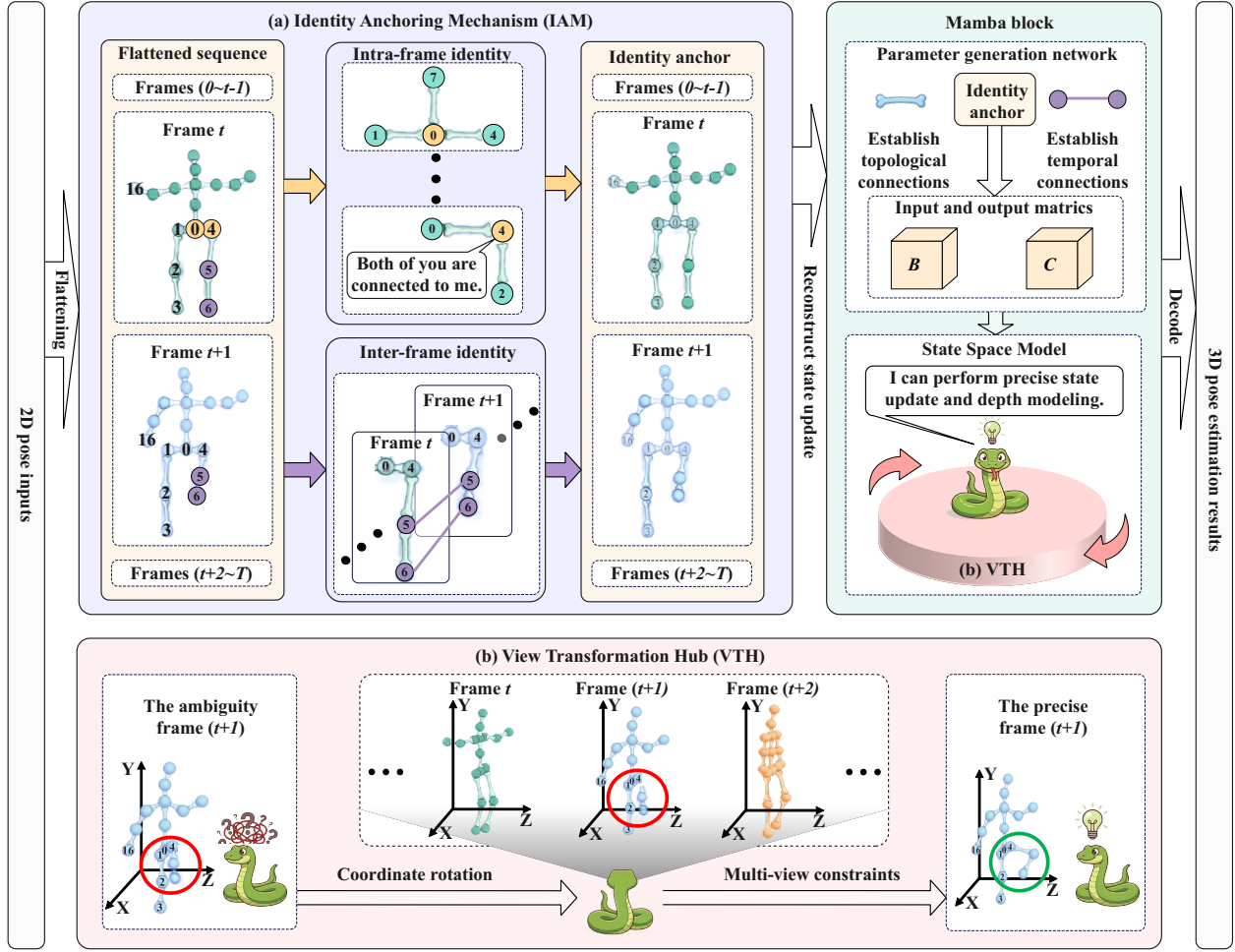


Figure 2. Overview of the StructMamPose. The framework comprises N blocks, each equipped with an IAM. The n -th block outputs an intermediate pose, which the VTH processes to obtain multi-view constraints. The resulting features are fed into subsequent blocks for further modeling. Finally, the outputs from the n -th and N -th blocks are fused to yield the final pose prediction.

and depth modeling capabilities. Finally, the intermediate and final block results are fused to estimate 3D poses.

3.3. The Identity Anchoring Mechanism (IAM)

The core challenge in applying Mamba to 3D-HPE stems from the conflict between its sequential modeling paradigm and the inherent structural topology of human pose data. Since Mamba’s state update matrices B and C rely on the flattened input data, the model fails to distinguish valid spatiotemporal correlations and spurious connections introduced by forcibly flattening, leading to confused state updates. To rectify this, we propose the IAM as shown in Fig.2 (a), which involves extracting spatiotemporal identities from the original 2D structure to encode precise semantics and correlations of joints. These identities are then injected into Mamba’s parameter generation network, which aims to anchor the selectivity of the state update matrices, enforcing update along valid temporal and topological dependencies.

Given the 2D pose input $P_{2D} \in \mathbb{R}^{T \times J \times 2}$, we first project it into the feature space to obtain the initial embedding $X \in \mathbb{R}^{T \times J \times C}$. This embedding is then flattened from horizontal-forward direction into a 1D sequence $F \in \mathbb{R}^{L \times C}$ ($L = T \times J$). To address the confusion caused by F , we employ a dual-branch feature encoding to reconstruct a spatiotemporal identity from X . First, based on the topological correlations of human joints, we construct a connectivity matrix $A \in \mathbb{R}^{T \times J \times J}$ for all nodes. We encode this into a joint-level identity $I_{\text{joint}} \in \mathbb{R}^{T \times J \times C}$ as follows:

$$I_{\text{joint}} = \text{Encode}(A). \quad (4)$$

This explicit definition encodes the authentic topological connections of each node and is critical for suppressing spurious connections between unrelated nodes. Next, we apply a GCN to each frame of X independently generate the frame-level identity $I_{\text{frame}} \in \mathbb{R}^{T \times J \times C}$ as:

$$I_{\text{frame}} = \text{GCN}(X), \quad (5)$$

where $\mathbf{I}_{\text{frame}}$ injects a frame-specific topological context into all nodes within each frame. This reinforces inter-frame feature discrepancy, enabling the model to clearly perceive contextual shifts during frame transitions. Subsequently, we flatten $\mathbf{I}_{\text{frame}}$ and $\mathbf{I}_{\text{joint}}$ and inject them into the generation network of the internal state matrices $\mathbf{B}$ and $\mathbf{C}$ as:

$$\mathbf{B}, \mathbf{C} = \text{Generation}(\mathbf{F} + \mathbf{I}_{\text{frame}} + \mathbf{I}_{\text{joint}}). \quad (6)$$

We utilize the generated $\mathbf{B}$ and $\mathbf{C}$ matrices to reconstruct the internal state updates of Mamba. Through spatial identity anchoring, these matrices update node states according to topological connections. Through temporal anchoring, these matrices break spurious dependencies among frames. Furthermore, the combination of spatial and temporal identities enables these matrices to anchor topologically correlated nodes within the frame, as well as the same node across different frames. This breaks the 'adjacency-implies-correlation' confusion within $\mathbf{F}$, compelling internal states to update strictly along authentic topological and temporal dependencies to achieve causally consistent modeling.

Based on the above, we reconstruct the internal state updates for the horizontal-forward flattening sequence. This effectively compensates for the inherent inability of horizontal flattening to incorporate valid temporal association. As the confusion present in the backward flattening sequence is consistent with the forward direction, we can directly apply the same identity to the backward sequence update. It is worth noting that we maintain the vertical flattening sequence in a standard processing. We believe that the primary confusion stems from the high-frequency unrelated correlations inherent in horizontal flattening. In contrast, vertical flattening preserves the continuity of joint motion trajectories, with discontinuities occurring only at joint switching boundaries ($J - 1$ times). Given that $J \ll L$, these temporal boundary effects are sparse. Considering the trade-off between performance and efficiency, we apt not to introduce additional parameters for this purpose. We only rely on the model's internal selective mechanism to implicitly suppress these sparse discontinuities.

3.4. The View Transformation Hub (VTH)

Relying on a single 2D view for 3D-HPE inherently leads to depth modeling confusion, as the absence of explicit spatial structural cues and multi-view constraints results in severe depth ambiguity. To overcome this limitation, we propose the VTH. Instead of relying on external multi-view data, VTH employs coordinate rotation to transform the original frontal view into a virtual side view within the internal feature space. It effectively converts the implicit depth dimension into explicit planar spatial observations. Through the complementary interaction between frontal and side views, the model explicitly perceives spatial structural cues. Furthermore, to rectify structural distortions in ambiguous

frames, we reference temporally correlated and unambiguous frames to impose multi-view constraints, thereby empowering the model with enhanced reasoning and rectifying capabilities for 3D human spatial structures.

As shown in Fig.2 (b), our framework comprises N stacked Mamba blocks, with the VTH embedded between the n -th and $(n + 1)$ -th blocks. In the n -th block, we decode the output feature $\mathbf{X}_n \in \mathbb{R}^{T \times J \times C}$ into an intermediate 3D pose result $\mathbf{P}_{3D,n} \in \mathbb{R}^{T \times J \times 3}$ from the frontal view. In this perspective, depth variations are invisible, forcing the model to rely on implicit inference based on ambiguous 2D poses to model depth information, leading to spatial modeling confusion and depth ambiguity. To resolve the confusion, we perform an internal coordinate rotation to transform $\mathbf{P}_{3D,n}$ into a virtual side-view pose $\mathbf{P}_{\text{rot},n} \in \mathbb{R}^{T \times J \times 3}$ as:

$$\mathbf{P}_{\text{rot},n} = \text{VTH}(\mathbf{P}_{3D,n}). \quad (7)$$

The core insight of this operation is that the original depth dimension is mapped to the planer dimension, thereby transforming the invisible and confused depth inference into observable side-view pose feature modeling. Subsequently, we re-encode $\mathbf{P}_{\text{rot},n}$ into side-view features $\mathbf{X}_{\text{side}} \in \mathbb{R}^{T \times J \times C}$. Ideally, combining features from both views enables the model to perceive the human spatial structure through multi-view modeling. However, depth-ambiguous poses in the original frontal view result in structural distortions in the generated side view after rotation. Directly combining such ambiguous views would further amplify errors. To rectify this error, we further reference temporally correlated frames to provide explicit spatiotemporal multi-view constraints. We utilize the original view features $\mathbf{X}_n$ as the query, and the side view features $\mathbf{X}_{\text{side}}$ as both the key and value to perform Cross-Attention $\text{CA}(\cdot)$ as:

$$\overline{\mathbf{X}}_n = \mathbf{X}_n + \text{CA}(\mathbf{X}_n, \mathbf{X}_{\text{side}}), \quad (8)$$

where $\overline{\mathbf{X}}_n \in \mathbb{R}^{T \times J \times C}$ denotes the fused features. This implies that the frontal view of each frame references both its corresponding side view and the side views of correlated, unambiguous frames, providing not only spatial structural cues but also spatiotemporal multi-view constraints. Then, $\overline{\mathbf{X}}_n$ is fed into subsequent blocks. This facilitates an evolution from planar feature extraction to spatial structure understanding, empowering the model to handle depth ambiguity. Finally, the output $\mathbf{X}_N \in \mathbb{R}^{T \times J \times C}$ from the final N -th block is decoded to yield the result $\mathbf{P}_{3D,N} \in \mathbb{R}^{T \times J \times 3}$. $\mathbf{P}_{3D,N}$ is fused with $\mathbf{P}_{3D,n}$ to obtain the final result $\mathbf{P}_{3D} \in \mathbb{R}^{T \times J \times 3}$. This strategy not only guarantees that the intermediate poses required for VTH rotation are supervised, but also facilitates a hierarchical cooperation between limbs structure and the body precision, enabling the model to achieve a coarse-to-fine refinement that ensures structure coherence and estimation accuracy.

Table 1. Quantitative comparisons on Human3.6M. T is the number of input frames. CE denotes estimating center frame only. MACs/frame represents multiply accumulate operations per output frame. P1: MPJPE. P2: P-MPJPE. $P1^\dagger$: P1 on 2D ground truth. The red and blue markers represent the best and second-best values.

Model size	Method	T	CE	Params(M)	MACs(G)	MACs/frame(M)	P1↓	P2↓	$P1^\dagger$ ↓
Small variants	P-STMO (Shan et al., 2022) (ECCV2022)	243	✓	6.2	0.7	3	42.8	34.4	29.3
	GLA-GCN (Yu et al., 2023) (ICCV2023)	243	✓	1.3	1.5	6	44.4	34.8	21.0
	HDFormer (Lan, 2024) (IJCAI2023)	96	×	3.7	0.6	6	42.6	33.1	21.6
	PoseMamba-S (Huang et al., 2025) (AAAI2025)	243	×	0.9	3.6	15	41.8	35.0	20.0
	StructMamPose-S	243	×	1.2	4.3	17	39.1	32.9	18.8
Base variants	STCFormer (Tang et al., 2023) (CVPR2023)	243	×	4.7	19.6	80	41.0	32.0	21.3
	PoseFormerV2 (Zhao et al., 2023) (CVPR2023)	243	✓	14.4	4.8	20	45.2	35.6	-
	PoseMamba-B (Huang et al., 2025) (AAAI2025)	243	×	3.4	13.9	57	40.8	34.3	16.8
	StructMamPose-B	243	×	4.9	15.1	62	38.1	32.3	14.2
Large variants	MixSTE (Zhang et al., 2022) (CVPR2022)	243	×	33.6	139.0	572	40.9	32.6	21.6
	STCFormer-L (Tang et al., 2023) (CVPR2023)	243	×	18.9	78.2	321	40.5	31.8	-
	MotionBERT (Zhu et al., 2023) (ICCV2023)	243	×	42.3	174.8	719	39.2	32.9	17.8
	MAGFormer-L (Mehraban et al., 2024) (WACV2024)	243	×	19.0	78.3	322	38.4	32.5	17.3
	KTPFormer (Peng et al., 2024) (CVPR2024)	243	×	35.2	76.1	313	40.1	31.9	19.0
	TCPFormer (Liu et al., 2025a) (AAAI2025)	243	✓	35.1	109.2	449	37.9	31.7	-
	PoseMagic (Zhang et al., 2025b) (AAAI2025)	243	×	14.4	20.3	84	37.5	-	-
	PoseMamba-X (Huang et al., 2025) (AAAI2025)	243	×	26.5	109.9	452	37.1	31.5	14.8
	StructMamPose-L	243	×	31.1	112.7	463	36.7	31.1	13.4

Table 2. Detailed MPJPE breakdown for all 15 actions on the Human3.6M dataset. The evaluation metric is P1 (mm).

Method	Venue	T	Dir.	Disc.	Eat	Greet	Phone	Photo	Pose	Pur.	Sit	SitD.	Smoke	Wait	WalkD.	Walk	WalkT.	Avg
MHFormer (Li et al., 2022)	CVPR'22	351	39.2	43.1	40.1	40.9	44.9	51.2	40.6	41.3	53.5	60.3	43.7	41.1	43.8	29.8	30.6	43.0
MixSTE (Zhang et al., 2022)	CVPR'22	243	37.6	40.9	37.3	39.7	42.3	49.9	40.1	39.8	51.7	55.0	42.1	39.8	41.0	27.9	27.9	40.9
P-STMO (Shan et al., 2022)	ECCV'22	243	38.9	42.7	40.4	41.1	45.6	49.7	40.9	39.9	55.5	59.4	44.9	42.2	42.7	29.4	29.4	42.8
STCFormer (Tang et al., 2023)	CVPR'23	243	39.6	41.6	37.4	38.8	43.1	51.1	39.1	39.7	51.4	57.4	41.8	38.5	40.7	27.1	28.6	41.0
GLA-GCN (Yu et al., 2023)	ICCV'23	243	41.3	44.3	40.8	41.8	45.9	54.1	42.1	41.5	57.8	62.9	45.0	42.8	45.9	29.4	29.9	44.4
KTPFormer (Peng et al., 2024)	CVPR'24	243	37.3	39.2	35.9	37.6	42.5	48.2	38.6	39.0	51.4	55.9	41.6	39.0	40.0	27.0	27.4	40.1
PoseRetNet (Mehraban et al., 2024)	ECCV'24	243	36.9	40.1	38.7	38.3	42.9	48.6	38.2	40.0	52.5	55.4	42.3	38.7	39.7	26.2	27.8	40.4
PoseMamba-S (Huang et al., 2025)	AAAI'25	243	38.9	43.4	39.2	35.5	44.6	53.6	38.9	38.2	55.7	60.1	43.1	38.7	39.0	28.4	27.9	41.8
SasMamba (Cui et al., 2026)	WACV'26	243	39.0	43.0	39.2	36.1	43.2	51.3	40.0	37.2	52.7	61.7	43.1	38.9	39.8	28.3	28.6	41.5
StructMamPose-S (Ours)	ICML'26	243	37.8	39.9	38.0	33.5	40.8	47.4	37.9	37.0	50.6	54.2	40.7	37.3	37.2	27.6	27.0	39.1

4. Experiments

4.1. Datasets and Evaluation Protocols

Human3.6M. As the most representative benchmark for 3D-HPE (Ionescu et al., 2013), this dataset comprises 3.6 million frames captured in a controlled indoor environment. It involves 11 professional actors performing 15 distinct daily activities such as sitting, walking, and direction, offering a wide range of complex poses. Following the standard configuration of PoseMamba (Huang et al., 2025), we employ subjects S1, S5, S6, S7, and S8 for training, while reserving subjects S9 and S11 for testing. Evaluation is conducted under two protocols: Protocol 1 reports the Mean Per Joint Position Error (MPJPE), measuring the Euclidean distance between predicted and ground truth joints after root alignment. Protocol 2 reports P-MPJPE (Procrustes MPJPE), which applies a rigid transformation for pose matching.

MPI-INF-3DHP. To evaluate the model’s generalization capabilities in complex scenarios, we employ this dataset (Mehta et al., 2017), which comprises 1.3 million frames. Distinguished from Human3.6M by its environmental diversity, it spans green-screen studios, realistic indoor settings, and dynamic outdoor scenes. Following standard practice, performance is assessed using three metrics: 3D PCK (Per-

centage of Correct Keypoints within a 150mm threshold), AUC (Area Under the Curve), and MPJPE.

4.2. Implementation Details

The proposed framework is implemented in PyTorch and executed on a single NVIDIA RTX 4090 GPU. To ensure a fair comparison, we align our loss objectives and data augmentation strategies with the classic baseline PoseMamba (Huang et al., 2025). The network parameters are optimized using the AdamW optimizer with a weight decay of 0.01 over a total of 120 training epochs. Regarding input data, we utilize the same 2D pose detection results as PoseMamba, supplemented by 2D ground truth annotations from both the Human3.6M and MPI-INF-3DHP datasets. Specifically for MPI-INF-3DHP experiments, we strictly adhere to the ground truth 2D configuration established in PoseMamba.

4.3. Comparison with State-of-the-art Methods

We provide three variants in different number of encoder blocks and channel dimension: StructMamPose-S, StructMamPose-B, and StructMamPose-L, for fair comparison with comparative models in different sizes.

Results on Human3.6M. We present a comprehensive eval-

Table 3. Quantitative comparisons on MPI-INF-3DHP dataset.

Method	T	PCK $\uparrow$	AUC $\uparrow$	MPJPE $\downarrow$
MixSTE (Zhang et al., 2022)	27	94.4	66.5	54.9
P-STMO (Shan et al., 2022)	81	97.9	75.8	32.2
GLA-GCN (Yu et al., 2023)	81	98.5	79.1	27.8
STCFormer (Tang et al., 2023)	81	98.7	83.9	23.1
MAGFormer (Mehraban et al., 2024)	81	98.2	85.3	16.2
KTPFormer (Peng et al., 2024)	81	98.9	85.9	16.7
TCPFormer (Liu et al., 2025a)	81	99.0	87.7	15.0
PoseMagic (Zhang et al., 2025b)	81	98.8	87.6	14.7
PoseMamba (Huang et al., 2025)	81	-	-	14.5
StructMamPose	81	98.9	88.2	14.1

Table 4. Ablation analysis of IAM and VTH. IAM-S and IAM-T denote the spatial and temporal identities, respectively.

Variants	IAM-S	IAM-T	VTH	P1 $\downarrow$
Remove all	×	×	×	41.8
Remove spatial identity	×	✓	✓	39.9
Remove temporal identity	✓	×	✓	40.0
Remove IAM	×	×	✓	40.7
Remove VTH	✓	✓	×	40.4
Remove none	✓	✓	✓	39.1

uation of these variants with recent methods in Table 1. All variants achieve significant improvements. Specifically, StructMamPose-S achieves the best P1, P2, and P1 † , reducing the errors by 2.7, 0.2, and 1.2 compared to the second-best methods PoseMamba-S (Huang et al., 2025) and HDFormer (Lan, 2024). Similarly, StructMamPose-B achieves the best performance in P1, P1 † , and is only surpassed by STCFormer (Tang et al., 2023) in P2. The reason is because P2 focuses more on the relative geometric relationships among joints. The global modeling of self-attention has a natural advantage. Importantly, as the model variant scales up, our StructMamPose-L surpasses STCFormer-L on P2, and achieves the best results with errors reduced by 0.4, 0.4, 1.4 compared to the second-best PoseMamba-X.

In addition, Table 2 provides an action-specific MPJPE breakdown on Human3.6M, revealing varying kinematic capabilities often obscured by average metrics. For complex folded postures with severe self-occlusions (e.g., "Photo", "SitDown"), standard 1D Mamba models often suffer from structural collapse. Here, StructMamPose-S achieves massive error reductions of 6.2 mm and 5.9 mm over PoseMamba-S. This leap is directly attributed to VTH exposing hidden depth errors as explicit planar distortions, and IAM anchoring topological identities. Conversely, while simple actions (e.g., "Walk") yield modest gains (0.8 mm) over already robust baselines, intricate hand-head interactions (e.g., "Phone", "Disc.") amplify error reductions to 3.8 mm and 3.5 mm. This confirms our state-of-the-art performance stems from conquering extreme monocular occlusions rather than merely saturating simple samples.

Results on Outdoor MPI-INF-3DHP. To ensure a fair comparison with methods using fixed configurations, we adjust the frame length. Specifically, we configure the input se-

quence length T to align with mainstream settings. The results summarized in Table 3 show that our model delivers superior performance. Except for being 0.1 lower than TCPFormer in PCK, our model achieves best results on all other metrics. The lower PCK is due to the strict thresholding favoring self-attention’s fine-grained perceptions, nevertheless, our performance remains highly competitive.

Accuracy vs. Efficiency Trade-off. As shown in Table 1, our method achieves a superior accuracy-efficiency trade-off. Although slightly larger in parameters than PoseMamba (while maintaining the same magnitude), our effective structural modeling delivers significantly superior accuracy across all scales. Furthermore, our Large variant sets a SOTA performance while requiring 26% fewer parameters and 35% fewer MACs than MotionBERT (Zhu et al., 2023). This demonstrates exceptional efficiency in modeling complex spatiotemporal features, providing a promising and efficient solution for 3D-HPE applications.

4.4. Ablation Study

We conduct a series of ablation studies on the Human3.6M dataset to validate the effectiveness of our framework and its core components, IAM and VTH. We select StructMamPose-S as the experimental model.

Quantitative Analysis of IAM and VTH. We conduct component-wise ablation studies as shown in Table 4. The baseline, devoid of both IAM and VTH, exhibits an error of 41.8 mm. Regarding IAM, ablating the spatial or temporal identity individually results in MPJPE of 39.9 mm and 40.0 mm, respectively, while removing both identities entirely increases the error to 40.7 mm. These degradations, compared to the full model (39.1 mm), highlight the importance of combining spatiotemporal identities. This performance gain is attributed to IAM’s capability to anchor precise spatiotemporal relationships for each node, thereby ensuring accurate state updates. On the other hand, removing VTH results in an MPJPE of 40.4 mm. The 1.3 mm performance drop verifies the efficacy of VTH. This improvement is credited to the explicit spatial structure and multi-view constraints provided by VTH, which enhance the network’s capability to capture human spatial structure for robust 3D-HPE.

Qualitative Analysis of IAM on Identity Anchoring. IAM aims to endow nodes with spatiotemporal identities to anchor internal state updates. To verify whether IAM effectively guides state updates according to the anchored identities, we compute the interaction map between the horizontal forward scanning matrices B and C . B governs the encoding of input information, while C controls the retrieval of states. Their product serves as a "Pseudo-Attention Map" to quantify the update dependency strength among joints. As shown in Figure 3, Node 3 is forced to be adjacent to the topologically unrelated Node 4 within the flattened se-

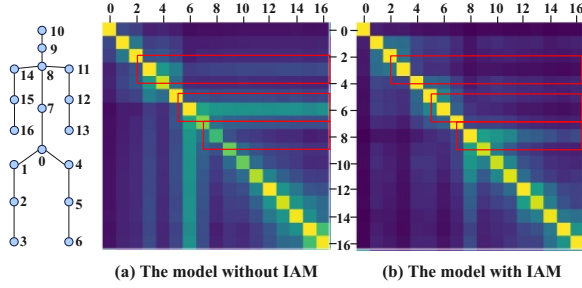


Figure 3. The interactive attention score maps of matrices B and C . IAM successfully resolves the update confusion in nodes 3 and 6, and establishes the correct topological update for node 8.

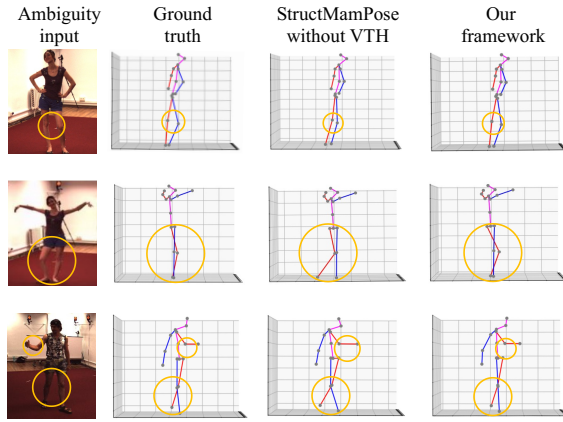


Figure 4. Depth estimation visualization. Our model achieves smaller errors in the depth ambiguity marked in yellow circles.

quence. In the model without IAM (a), Node 3 establishes high responses with all subsequent nodes, which violates the human body topology. In the model with IAM (b), Node 3 correctly breaks the connection with Node 4 and further avoids connecting to unrelated subsequent nodes. We attribute this to the fact that the model without IAM relies only on the flattened sequence for state updates, which forces a relationship between Node 3 and Node 4. This contradiction of the true topology causes update confusion, modeling erroneous dependencies with all subsequent nodes. On the other hand, Node 8 should topologically relate to all subsequent nodes. The model with IAM correctly establishes these connections, whereas the model without IAM ignores them. The reason is that the flattened sequence misleadingly suggests to matrices B and C that Node 8 is adjacent only to Node 9. Therefore, the success of IAM lies in injecting spatiotemporal identities into B and C , allowing the model to anchor relevant joints and updating correct dependencies based on the topological relationships.

Qualitative Analysis of VTH on Depth Ambiguity. The core objective of VTH is to enhance the network’s spatial

Table 5. Ablation analysis of VTH insertion layer. The error decreases as the insertion layer depth n increases.

Insertion depth n	P1 ↓	P2 ↓	P1† ↓
$n = 1$	41.2	34.4	19.7
$n = 3$	40.9	34.3	19.6
$n = 5$	40.6	34.1	19.3
$n = 7$	39.7	33.2	19.1
$n = 9$	39.1	32.9	18.8

understanding capability and alleviate the depth ambiguity. To validate the effectiveness of VTH, we select several challenging samples with ambiguous 2D inputs from the Human3.6M and visualize the 3D pose estimation results along the depth axis. As illustrated in Figure 4, when VTH is removed, the model exhibits significant deviation from the ground truth in yellow circles, resulting in incorrect overall pose predictions. In contrast, although our method exhibits minor deviations, the predicted poses remain consistent with the ground truth. This validates the effectiveness of VTH. This success is attributed to the provision of explicit spatial cues and multi-view constraints, which facilitate depth feature extraction and enhance the network’s capability for spatial understanding and error correction.

Impact of VTH Insertion Depth and Rotation Angles.

The VTH is integrated between the n -th and $(n+1)$ th layers. To investigate the optimal layers, we evaluate five variants with $n \in \{1, 3, 5, 7, 9\}$. As shown in Table 5, the variant with $n = 9$ yields superior performance with 39.1 mm, while others exhibit a gradual increase in error as the layer n decreases. This suggests that VTH at later stage of the framework provides the most effective multi-view constraints. We attribute this to the feature representation. The early layers of the network tend to extract local features, which are limited to representing localized limb details and lack global context. In contrast, deep layers encode rich high-level semantics and body structures. Consequently, rotating the coordinates in the deeper layers generates side-views that more accurately describe the human pose, providing robust spatial cues and constraints for the subsequent layers.

Furthermore, to verify the specific effect of internal coordinate rotation angles on alleviating depth ambiguity, we evaluated the performance of rotating the intermediate 3D coordinates around the Y-axis by different angles ($0^\circ, 30^\circ, 45^\circ, 60^\circ, 90^\circ$) on StructMamPose-S. Note that setting the angle to 0° is equivalent to removing the rotation. As shown in Table 6, the 90° orthogonal rotation achieves the maximum performance gain (an error reduction of 1.3 mm compared to 0°). Physically, a 90° rotation completely swaps the depth axis with the horizontal axis. This maximizes the exposure of implicit depth errors by transforming them into explicit 2D planar distortions. We further evaluated a multi-angle fusion strategy (e.g., $90^\circ + 60^\circ$) and found no performance improvement. This proves that the orthogo-

Table 6. Ablation study on the internal coordinate rotation angles of the VTH module (evaluated on StructMamPose-S). The evaluation metric is MPJPE (P1, mm).

Rotation Angle (°)	0° (None)	30°	45°	60°	90°	90° + 60°
StructMamPose-S ↓	40.4	40.1	40.0	39.7	39.1	39.3

Table 7. Robustness evaluation in challenging motion and occlusion conditions on Human3.6M. ‘Overall’ denotes the performance on the entire test set. The evaluation metric is P1.

Method	Overall ↓	Easy ↓	Med. ↓	Hard ↓	Low ↓	High ↓
DiffPose	37.0	32.8	42.8	54.4	35.0	42.9
PoseMagic	37.5	33.1	43.6	56.2	35.5	43.5
PoseMamba-X	37.1	32.8	43.1	55.3	35.1	43.2
StructMamPose-L	36.7	32.7	42.5	53.1	34.9	42.2

nal virtual side-view already provides sufficient structural constraints, and introducing additional intermediate views conversely injects redundant noise.

Robustness in Challenging Scenarios. The true robustness of a 3D-HPE framework is determined by its performance under demanding conditions. Because standard test sets are dominated by simple samples, the actual capability of a model to handle adverse situations is frequently obscured. We hypothesize that by resolving internal state update confusion and depth ambiguity through the IAM and VTH, our StructMamPose will demonstrate substantial performance advantages over standard Mamba baselines, particularly under severe occlusion and intense motion. To demonstrate this, we apply K-Means clustering to re-partition the Human3.6M test set along two crucial dimensions. Regarding the spatial occlusion dimension, we cluster the data based on 2D joint confidence, forming Easy, Medium, and Hard occlusion subsets (with an approximate ratio of 7:2:1). Simultaneously, we cluster temporal motions into Low and High intensity subsets using kinematic vectors incorporating velocity, acceleration, and FFT (approximate ratio 3:1). As presented in Table 7, whereas other approaches experience severe performance drops under these challenging conditions, StructMamPose-L preserves remarkable stability. Specifically, on the Hard occlusion subset, our model reduces the error from 55.3 to 53.1 mm, achieving a substantial absolute improvement of 2.2 mm over PoseMamba-X. In contrast, the improvement on the saturated Easy subset is merely 0.1 mm (from 32.8 to 32.7). Furthermore, it achieves a 2.3% relative error reduction on the High motion intensity subset. These targeted gains exceed the overall average improvement of 0.4 mm. Qualitatively, Figure 5 further demonstrates this robustness; compared to the PoseMamba baseline, our method reconstructs challenging body parts (yellow circles) with higher fidelity to the ground truth under severe occlusion and unconventional poses. This confirms that explicitly introducing multi-view constraints and anchoring spatiotemporal identities effectively prevents the structural distortions caused by monocular depth ambiguity and forced 1D sequence flattening.

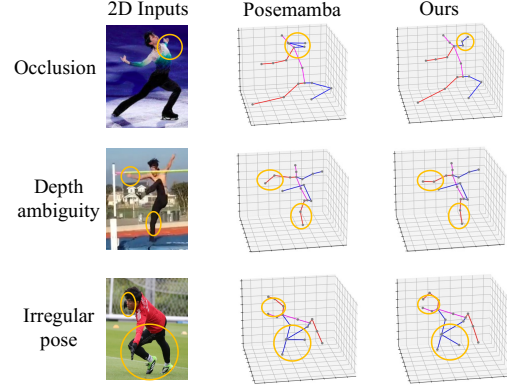


Figure 5. Generalization Comparison. Our method achieves superior results on challenging parts marked in yellow circles.

5. Limitations and Future Works

Although IAM and VTH achieve excellent performance, the current framework remains confined to the visual domain. When addressing intelligent tasks such as human-machine interaction and human referring, effective execution necessitates not only spatial perception but also cross-modal semantic alignment to interpret natural language. Our future works will explore these limitations, endowing the model with the ability to comprehend semantic cues, bridging the gap between spatial estimation and intelligent interaction.

6. Conclusion

We propose StructMamPose with IAM and VTH to robust 3D-HPE. The IAM anchors joint relationships to update internal state, while the VTH performs coordinates rotation to provide spatiotemporal multi-view constraints. These components are integrated to reconstruct precise internal state update and depth modeling, enabling a transition from simple coordinate regression to comprehensive structure understanding. Extensive experiments demonstrate excellent performance. Our work establishes a structure-aware Mamba methodology for future 3D-HPE research.

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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Appendix

A. Impact of Multiple VTH Insertions

In our standard design, a single VTH module is inserted at a late stage ($n = 9$). To explore whether multiple insertions yield further gains, we evaluated two strategies: (1) sequential insertion at the same location ($n = 9 \& 9$), and (2) dispersed insertions across different layers ($n = 3 \& 9$, $n = 5 \& 9$, and $n = 7 \& 9$).

As shown in Table 8, inserting two sequential VTH modules at $n = 9$ yields a marginal 0.03 mm MPJPE improvement over the baseline. Similarly, inserting VTH at intermediate layers alongside the final layer improves accuracy by just 0.01, 0.02, and 0.05 mm, respectively. These results demonstrate that a single late-stage VTH sufficiently corrects depth ambiguity. Additional insertions only introduce extra parameters for negligible gains. Weighing this trade-off, a single insertion remains the optimal choice.

Table 8. Ablation study on multiple insertions of the VTH module. The baseline uses a single late-stage insertion at $n = 9$. The evaluation metric is MPJPE (mm).

Insertion Strategy	Insertion Layers	MPJPE ↓
Single Insertion (Baseline)	$n = 9$	39.10
Sequential Insertion	$n = 9 \& 9$	39.07 (-0.03)
Dispersed Insertion	$n = 3 \& 9$	39.09 (-0.01)
	$n = 5 \& 9$	39.08 (-0.02)
	$n = 7 \& 9$	39.05 (-0.05)

B. Ablation on Deeper VTH Insertion Layers

Following the discussions in Table 5 of the main text, where we investigated the impact of VTH insertion depth and established that increasing the depth from $n = 1$ to $n = 9$ monotonically reduces estimation errors on StructMamPose-S, it is verified that late-stage features provide more accurate high-level semantics for virtual view generation. To further investigate whether this performance gain extends to the ultimate layers of a larger network and to identify the exact inflection point for feature fusion, we evaluate even deeper insertion layers ($n \in \{13, 15, 17, 19, 20\}$) on StructMamPose-L.

As shown in Table 9, when migrating the VTH module into these late stages of the 20-block architecture, the MPJPE exhibits a first-decreasing-then-increasing trend, achieving the optimal performance of 36.7 mm at the 17th layer. This confirms that VTH placement must strike a precise balance between feature representation and subsequent processing. While the module requires a sufficiently deep layer (*e.g.*, Layer 17) to capture stable global body structures, the framework must crucially reserve the final few blocks (*e.g.*, Layers 18–20) to fully digest and fuse the generated multi-view constraints before the final 3D pose regression.

Table 9. Ablation study on deeper VTH insertion layers evaluated on StructMamPose-L. The evaluation metric is MPJPE (mm).

Insertion Layer (n)	13	15	17	19	20
StructMamPose-L ↓	37.5	37.1	36.7	37.0	37.4

C. Failure Cases

While StructMamPose demonstrates strong robustness across standard scenarios, its structural reasoning mechanisms naturally encounter limits when confronting extreme out-of-distribution (OOD) poses. As shown in Figure 6, we identify two primary failure scenarios: First, VTH relies on intermediate 3D predictions for view transformation. For highly unconventional actions (*e.g.*, severe forward bending or unnatural contortions), the model struggles to output a reliable intermediate geometry. Rotating these corrupted poses by 90° generates anatomically irrational virtual side-views. If the temporal neighbors are also undergoing similar extreme motions, the cross-attention mechanism fails to retrieve reliable geometric cues, causing the endogenous correction to break down. Second, extreme limb folding directly compromises the front-end 2D detector, resulting in highly entangled and noisy joint coordinates. Although the Identity Anchoring Mechanism (IAM) is designed to prevent structural distortions via topological anchors, this fundamental collapse in 2D observations (semantic aliasing) occasionally exceeds the correction limits of the deep Mamba blocks.

These observations suggest that while explicit structural reasoning effectively mitigates depth ambiguity, future iterations could bypass the inherent vulnerability of 2D representations under extreme occlusion by tightly coupling with raw visual features.

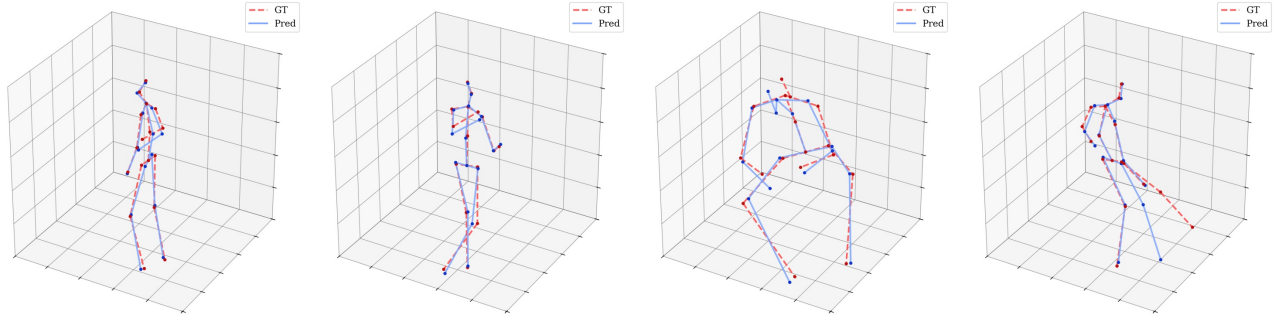


Figure 6. Visual analysis of StructMamPose under challenging failure scenarios. In cases involving out-of-distribution spatial orientations and extreme limb folding, the model encounters challenges in perfectly capturing the highly unnatural geometric structures, occasionally leading to depth and orientation misalignments.